Homework 2

Analysis

Your Name Here

This homework consists of two parts: the first is an extended data analysis similar to Homework 1; the second is a brief simulation study to help you distinguish concepts related to simulation studies.

Part 1: SEDA data set

Background

Gender achievement gaps in education have been well-documented over the years – studies consistently find boys outperforming girls on math tests and girls outperforming boys on reading and language tests. A particularly controversial article was published in Science in 1980 arguing that this pattern was due to an 'innate' difference in ability (focusing on mathematics rather than on reading and language). Such views persisted in part because studying systematic patterns in achievement nationwide was a challenge due to differential testing standards across school districts and the general lack of availability of large-scale data.

It is only recently that data-driven research has begun to reveal socioeconomic drivers of achievement gaps. The Standford Educational Data Archive (SEDA)—also known as the Educational Opportunity Project at Stanford University—is a publicly available database on academic achievement and educational opportunity in U.S. schools that has played a key role in this effort. The database is part of a broader initiave aiming to improve educational opportunity by enabling researchers and policymakers to identify systemic drivers of disparity.

> SEDA provides detailed data on educational conditions, contexts, and outcomes in school districts and counties across the United States. It includes measures of academic achievement and achievement gaps, as well as district-level indicators of racial and socioeconomic composition, racial and socioeconomic segregation, and other features of the schooling system.

The database standardizes average test scores for more than 10k U.S. school districts relative to national standards. This allows comparisons across districts, grade levels, and years. The test score data are drawn from the U.S. Department of Education. In addition, multiple data sources—including the American Community Survey and the Common Core of Data—are integrated to provide district-level socioeconomic and demographic information.

A 2018 study using SEDA data identified the following persistent patterns across grade levels 3 - 8 and school years from 2008 through 2015:

- A consistent reading and language achievement gap favoring girls;
- *No* national math achievement gap on average; and
- Local math achievement gaps that depend on the socioeconomic conditions of school districts.

You can read about the main findings of the study in this brief NY Times article.

Below, we'll work with selected portions of the database. The full datasets can be downloaded here.

Assignment objectives

In this assignment, you'll explore achievement gaps in California school districts in 2018, reproducing the findings described in the article above on a more local scale. You'll practice the following:

- Review of data documentation
- Assessment of sampling design and scope of inference
- Data tidying operations
  - Slicing and filtering
  - Merging multiple data frames
  - Pivoting tables
  - Renaming and reordering variables
- Constructing exploratory graphics and visualizing trends
- Data aggregations
- Narrative summary of exploratory analysis

Import and assessment of datasets

You'll work with test data and socioeconomic covariates aggregated to the school district level. These data are stored in two separate tables. Here you'll examine them and review data documentation.

Test score data

The first few rows of the test data are shown below.

```
ca_main <- read_csv("data/ca-main.csv")
```

```
Rows: 7513 Columns: 59
-- Column specification --------------------------------------------------------
Delimiter: ","
chr  (3): stateabb, sedaleaname, subject
dbl (52): sedalea, grade, cs_mn_all, cs_mnse_all, totgyb_all, cs_mn_asn, cs_...
lgl  (4): cs_mn_nam, cs_mnse_nam, cs_mn_wng, cs_mnse_wng

i Use `spec()` to retrieve the full column specification for this data.
i Specify the column types or set `show_col_types = FALSE` to quiet this message.
```

```
ca_main |> head(n=5)
```

```
# A tibble: 5 x 59
  sedalea grade stateabb sedaleaname    subject cs_mn_all cs_mnse_all totgyb_all
    <dbl> <dbl> <chr>    <chr>          <chr>       <dbl>       <dbl>      <dbl>
1  600001     4 CA       ACTON-AGUA DU~ mth      -3.67e-1      0.109          86
2  600001     4 CA       ACTON-AGUA DU~ rla       5.69e-3      0.117          85
3  600001     6 CA       ACTON-AGUA DU~ rla      -4.02e-5      0.0922        114
4  600001     8 CA       ACTON-AGUA DU~ mth      -9.77e-2      0.103          98
5  600001     8 CA       ACTON-AGUA DU~ rla      -4.86e-1      0.111          99
# i 51 more variables: cs_mn_asn <dbl>, cs_mnse_asn <dbl>, totgyb_asn <dbl>,
#   cs_mn_blk <dbl>, cs_mnse_blk <dbl>, totgyb_blk <dbl>, cs_mn_ecd <dbl>,
#   cs_mnse_ecd <dbl>, totgyb_ecd <dbl>, cs_mn_fem <dbl>, cs_mnse_fem <dbl>,
#   totgyb_fem <dbl>, cs_mn_hsp <dbl>, cs_mnse_hsp <dbl>, totgyb_hsp <dbl>,
#   cs_mn_mal <dbl>, cs_mnse_mal <dbl>, totgyb_mal <dbl>, cs_mn_mfg <dbl>,
#   cs_mnse_mfg <dbl>, totgyb_mfg <dbl>, cs_mn_mtr <dbl>, cs_mnse_mtr <dbl>,
#   totgyb_mtr <dbl>, cs_mn_nam <lgl>, cs_mnse_nam <lgl>, totgyb_nam <dbl>, ...
```

Column name	Meaning
`sedalea`	District ID
`grade`	Grade level
`stateabb`	State abbreviation
`sedaleaname`	District name
`subject`	Test subject
`cs_mn_...`	Estimated mean test score
`cs_mnse_...`	Standard error for estimated mean test score
`totgyb_...`	Number of individual tests used to estimate the mean score

The test score means for each district are named `cs_mn_...` with an abbreviation indicating subgroup (such as mean score for all `cs_mean_all`, for boys `cs_mean_mal`, for white students `cs_mn_wht`, and so on). Notice that these are generally small-ish: decimal numbers between -0.5 and 0.5.

These means are *estimated* from a number of individual student tests and *standardized* relative to national averages. They represent the number of standard deviations by which a district mean differs from the national average. So, for instance, the value `cs_mn_all = 0.1` indicates that the district average is estimated to be 0.1 standard deviations greater than the national average on the corresponding test and at the corresponding grade level.

Question 1: Interpreting test score values

Interpret the average math test score for all 4th grade students in Acton-Agua Dulce Unified School District (the first row of the dataset shown above).

Type your answer here, replacing this text.

Covariate data

The first few rows of the covariate data are shown below.

```
ca_cov <- read_csv("data/ca-cov.csv")
```

```
Rows: 4340 Columns: 60
-- Column specification --------------------------------------------------------
Delimiter: ","
chr  (3): sedaleanm, locale, gslo
dbl (57): sedalea, grade, urban, suburb, town, rural, perind, perasn, perhsp...
```

```
i Use `spec()` to retrieve the full column specification for this data.
i Specify the column types or set `show_col_types = FALSE` to quiet this message.
```

```
ca_cov |> head(n = 5)
```

```
# A tibble: 5 x 60
  sedalea grade sedaleanm  urban suburb  town rural locale  perind perasn perhsp
    <dbl> <dbl> <chr>      <dbl>  <dbl> <dbl> <dbl> <chr>    <dbl>  <dbl>  <dbl>
1  600001     4 ACTON-AGU~     0      0     0     1 Rural~ 0.00389 0.0459  0.308
2  600001     5 ACTON-AGU~     0      0     0     1 Rural~ 0.00379 0.0467  0.304
3  600001     6 ACTON-AGU~     0      0     0     1 Rural~ 0.00322 0.0437  0.309
4  600001     7 ACTON-AGU~     0      0     0     1 Rural~ 0.00341 0.0441  0.291
5  600001     8 ACTON-AGU~     0      0     0     1 Rural~ 0.00379 0.0453  0.309
# i 49 more variables: perblk <dbl>, perwht <dbl>, perfl <dbl>, perrl <dbl>,
#   perecd <dbl>, perell <dbl>, perspeced <dbl>, totenrl <dbl>, gslo <chr>,
#   gshi <dbl>, hswhtblk <dbl>, hswhthsp <dbl>, hsflnfl <dbl>, hsecdnec <dbl>,
#   rswhtblk <dbl>, rswhthsp <dbl>, rsflnfl <dbl>, rsecdnec <dbl>,
#   perfrl <dbl>, sesall <dbl>, sesblk <dbl>, seshsp <dbl>, seswht <dbl>,
#   lninc50all <dbl>, lninc50blk <dbl>, lninc50hsp <dbl>, lninc50wht <dbl>,
#   baplusall <dbl>, baplusblk <dbl>, baplushsp <dbl>, bapluswht <dbl>, ...
```

The column information is as follows:

Column name	Meaning
`sedalea`	District ID
`grade`	Grade level
`sedaleanm`	District name
`urban`	Indicator: is the district in an urban locale?
`suburb`	Indicator: is the district in a suburban locale?
`town`	Indicator: is the district in a town locale?
`rural`	Indicator: is the district in a rural locale?
`locale`	Description of district locale
Remaining variables	Demographic and socioeconomic measures

You will only be working with a handful of the demographic and socioeconomic measures, so you can put off getting acquainted with those until selecting a subset of variables.

Question 2: Data semantics

In the non-public data, observational units are students—test scores are measured for each student. However, in the SEDA data you've imported, scores are aggregated values. ASSUME that the `ca_main` table has one observation per row (i.e., it is already tidy), what is the observational unit? Is it one district? One grade in one district? Or something else.

Are the units the same or different for the covariate table `ca_cov`, assuming it is also one observation per row?

Type your answer here, replacing this text.

Question 3: Sample sizes

How many observations are in each dataset? Count the number of units in the test dataset and the number of units in the covariate dataset separately. Store the values as `ca_main_units` and `ca_cov_units`, respectively. It is tempting to use `nrow()`, but how do we know for sure that there is no replicated observations (multiple rows corresponding to the same unit)? (*Hint*: use `distinct()`)

```
# Count unique districts in each dataset
ca_main_units <- ...

ca_cov_units <- ...

print(paste('units in test score data: ', ca_main_units))
print(paste('units in covariate data: ', ca_cov_units))
```

Question 4: Sample characteristics and scope of inference

Answer the questions below about the sampling design in a short paragraph. You do not need to dig through any data documentation in order to resolve these questions.

- (i) What is the relevant population for the datasets you've imported?
- (ii) About what proportion (to within 0.1) of the population is captured in the sample? (*Hint*: have a look at this website.)
- (iii) Considering that the sampling frame is not identified clearly, what kind of dataset do you suspect this is (*e.g.*, administrative, data from a 'typical sample', census, etc.)?
- (iv) In light of your description of the sample characteristics, what is the scope of inference for this dataset?

Type your answer here, replacing this text.

Data tidying

You'll work with the following variables from each dataset:

- **Test score data**
  - District ID
  - District name
  - Grade
  - Test subject
  - Estimated male score (*Hint*: `cs_mn_mal`)
  - Estimated female score
- **Covariate data**
  - District ID
  - Locale
  - Grade
  - Socioeconomic index (all demographic groups) (*Hint*: try `ses`)
  - Log median income (all demographic groups)
  - Poverty rate (all demographic groups)
  - Unemployment rate (all demographic groups)
  - SNAP benefit receipt rate (all demographic groups)

Question 5: Variable names of interest

Download the codebooks by opening the 'data' directory and downloading the codebook files. Identify the variables listed above, and store the column names in lists named `main_vars` and `cov_vars`.

```
# Store variable names of interest
main_vars <- ...

cov_vars <- ...
```

Question 6: Slice columns

Use your result from above to slice the columns of interest from the test score and covariate data. Store the resulting data frames as `main_sub` and `cov_sub`.

```
# Slice columns to select variables of interest
main_sub <- ...

cov_sub <- ...
```

Question 7: Merge

Merge the covariate and test score data on both the ***district ID*** and ***grade level*** columns, and retain all the columns from the test score data (meaning, merge the covariate data *to* the test score data). You should use the `left_join()` with `main_sub` as the main table. Store the resulting data frame as `rawdata` and print the first four rows. (We need to learn how to join by two columns).

```
rawdata <- ...
```

Question 8: Rename and reorder columns

Use `mutate` to create a new variable called `gender_gap` which is `cs_mn_mal - cs_mn_fem`. Then rename and rearrange the columns of `rawdata` so that they appear in the following order and with the following names:

- `district_id`, `District`, `Locale`, `log_median_income`, `poverty_rate`, `unemployment_rate`, `snap_rate`, `socioeconomic_index`, `Grade`, `Subject`, `gender_gap`.

Notice that none of these names contains a space. Names with spaces have to be wrapped in backticks (`` `like this` ``) every single time you refer to them; names joined with underscores do not. Multi-word names are snake_case for that reason, and single-word names are left capitalized. You will feel the difference in this homework.

Select only the columns above, store the resulting data frame as `rawdata_mod1` and print the first four rows.

```
# Define dictionary for renaming columns
name_dict <- ...

# Specify column order
col_order <- ...

# Rename and reorder and calculate gender gap
rawdata_mod1 <- ...
```

```
# Print first four rows
...
```

Question 9: Pivot

Notice that the `gender_gap` column contains the values of two variables: the gap in estimated mean test scores for math tests, and the gap in estimated mean test scores for reading and language tests. To put the data in tidy format, use `pivot_wider` to pivot the table so that the gender gap column is spread into two columns `mth` and `rla`. Rename these columns `Math` and `Reading`. Store the result as `seda_data` and print the first four rows.

```
# Pivot to unstack gender gap
seda_data <- ...

# Print first four rows
...
```

Your final dataset should have an identical variable structure as the dataframe below. You can use this to check your answer and revise any portions above that lead to different results.

```
# intended result
data_reference <- read_csv('data/tidy-seda-check.csv')
```

```
Rows: 10 Columns: 11
-- Column specification --------------------------------------------------------
Delimiter: ","
chr (2): District, Locale
dbl (9): district_id, log_median_income, poverty_rate, unemployment_rate, sn...

i Use `spec()` to retrieve the full column specification for this data.
i Specify the column types or set `show_col_types = FALSE` to quiet this message.
```

```
data_reference
```

```
# A tibble: 10 x 11
   district_id District  Locale log_median_income poverty_rate unemployment_rate
         <dbl> <chr>     <chr>              <dbl>        <dbl>             <dbl>
 1      600001 ACTON-AG~ Rural~              11.4       0.0919            0.0489
 2      600001 ACTON-AG~ Rural~              11.4       0.0919            0.0489
 3      600001 ACTON-AG~ Rural~              11.4       0.0919            0.0489
```

```
 4      600006 ROSS VAL~ Subur~              11.6       0.0414            0.0483
 5      600006 ROSS VAL~ Subur~              11.6       0.0414            0.0483
 6      600006 ROSS VAL~ Subur~              11.6       0.0414            0.0483
 7      600006 ROSS VAL~ Subur~              11.6       0.0414            0.0483
 8      600011 FORT SAG~ Rural~              10.7       0.160             0.0663
 9      600011 FORT SAG~ Rural~              10.7       0.160             0.0663
10      600011 FORT SAG~ Rural~              10.7       0.160             0.0663
# i 5 more variables: snap_rate <dbl>, socioeconomic_index <dbl>, Grade <dbl>,
#   Math <dbl>, Reading <dbl>
```

Question 10: Missing values

Gap estimates were not calculated for certain grades in certain districts due to small sample sizes (not enough individual tests recorded). Answer the following:

- (i) What proportion of rows are missing for each of the reading and math gap variables? Store these values as `math_missing` and `reading_missing`, respectively.
- (ii) What proportion of *districts* (not rows!) have missing gap estimates for at least one test subject for at least one grade level? Store the value as `district_missing`.

```
# proportion of missing values
math_missing <- ...
reading_missing <- ...

# proportion of districts with missing values
district_missing <- ...
```

Question 11: Missing mechanism

Do you notice that missing values tend to occur more frequently in certain districts than in others? How might simply dropping all districts with any missing data harm our analysis?

Type your answer here, replacing this text.

Question 12: Santa Barbara Unified

It's often helpful to build intuition and check your data by exploring observations for which you are (or might be familiar). Filter the district to "SANTA BARBARA UNIFIED", the district for the city of Santa Barbara. schools. Select `Grade`, `Math` and `Reading`. Print all rows of the data frame for these 3 columns only. For which grades and subjects did boys outperform girls in Santa Barbara?

Optional: if you went to elementary school or Junior High in California, complete the exercise for your former school district instead.

```
...
```

Exploratory graphics

For the purpose of visualizing the relationship between estimated gender gaps and socioeconomic variables, you'll find it more helpful to work with a longer *non-tidy* version of the data. The cell below rearranges the dataset so that one column contains an estimated gap, one column contains the value of a socioeconomic variable, and the remaining columns record the gap type and variable identity.

```
plot_df <- seda_data |>
  pivot_longer(cols = c(Math, Reading),
               names_to = "Subject",
               values_to = "Gap") |>
  pivot_longer(cols = log_median_income:socioeconomic_index,
               names_to = "socioeconomic_type",
               values_to = "socioeconomic_measure")
# preview
head(plot_df)
```

Gender gaps and socioeconomic factors

Question 13: Scatter Plot

Create a panel of scatter plots showing the relationship between estimated gender gap and socioeconomic factors, with points colored by test subject. Use `facet_wrap` to show a plot for each of the 5 socioeconimc variables. Adjust the size and transparency of the points to reduce the impact of overplotting. Which subject has the larger gap, reading or math? Which gender is performing better in this subject?

Hint: use `scales="free_x"` in the `facet_wrap` call to ensure that each faceted plot gets it's own x-limits.

```
fig1 <- ...

fig1
```

Type your answer here, replacing this text

Question 14: linear fits.

You can tell from the previous plot which subject has the larger gender gap and who is outperforming, but it's hard to distinguish more subtle patterns about how these relationships depend on socioeconomic variables. Instead of a scatter plot, we'll use `geom_smooth` which can be used to to fit smooth functions to the data. We'll focus on linear fits to the data by specifying the `geom_smooth(method="lm")`, where here `lm` stands for "linear model". Again, color the lines by subject (setting the `col` aesthetic) and facet by socioeconomic variable (and set `scale="free_x"` again in the `facet_call`).

For what subjects and socioeconomic values do boys outperform girls? Is the relationship between socioeconomic variables and gender gap the same for each subject?

Type your answer here, replacing this text

```
fig2 <- ...

fig2
```

Question 15: Relationships by grade level

Modify the plot above to show these relationships by grade level: generate a panel of scatterplots of gap against socioeconomic measures by subject (which defines `linetype`), coloring the lines according to `Grade`.

Since `Grade` is a numeric variable by default, if we want separate colored lines for each grade, we need to specify that `Grade` should be treated as factor in the aesthetics. To do so, set `col=as.factor(Grade)`.

After your first pass, try adding `colorspace::scale_color_discrete_sequential(palette="Viridis")` to your `ggplot` to set the grades to the Viridis color palette.

Does the pattern shown in the plot above persist within each grade level? For which grades are the gender gaps the largest? Is this pattern consistent across subjects?

```
fig3 <- ...

fig3
```

Type your answer here, replacing this text.

Question 16: Does locale matter?

Let's focus just on math scores and consider whether or not local matters. The chunk below adds a new variable, `locale_coarse` which is a coarser summary of locale which includes just "Rural", "Town", "Suburb" and "City", but not the more detailed descriptions.

(i) What are the all possible values that `plot_df$Locale` may take? Did you notice a typo in the data?

```
...
```

(ii) Create the new variable `locale_coarse`. You may need `case_when()` and a regular expression function.

```
plot_df <- plot_df |> mutate(locale_coarse =
                    # create new variable locale_coarse here
                    )
```

(iii) As you did above, make a plot showing the relationship between socioeconomic variables and the math gap. Do so by first filtering `plot_df` to only include math subjects and then use `geom_smooth(method="lm", se=FALSE)`. Set the color aesthetic to `locale_coarse` so that a line is produced for each locale. Use any color palette you'd like for the lines. Does any locales have larger average gaps than others? Do some locales show a stronger or weaker relationship with the socioeconomic variables?

```
...
```

Type your answer here in 2-3 sentences, replacing this text

Question 17: Aggregation across grade levels

Compute the mean estimated achievement gap in each subject, averaged across grade levels to the `District` level. We also want to retain the district-level socioeconomic variables. Store the resulting data frame as `seda_data_agg`.

Note: best practice here would be to aggregate just the test scores by district and then re-merge the result with the district-level socioeconomic variables. However, since the district-level socioeconomic variables do not differ by grade within a district, averaging them across grade levels by district together with the test scores will simply return their original values; so the aggregation can be applied across *all* columns for a fast-and-loose way to obtain the desired result.

```
# aggregate across grades
seda_data_agg <- ...
```

The cell below adds an `income_bracket` variable by cutting the median income into 8 contiguous intervals using `cut()`.

```
## Filter nans
seda_data_agg <- seda_data_agg |>
  drop_na(Math, Reading, log_median_income) |>
  mutate(
    income_bracket = cut(
      exp(log_median_income),
      breaks = 8))
```

As an example, the next cell tabulates the average socioeconomic measures and estimated gaps across districts by income bracket. Does the data in this table roughly match the trends in the plots you saw above?

```
seda_data_agg |>
  group_by(income_bracket) |>
  summarise(
    across(-c(district_id, District, Locale, log_median_income), # The minus indicates "not c
      \(x) mean(x, na.rm=TRUE)
    )
  ) |>
  arrange(desc(income_bracket)) |> ## arrange from highest income to lowest
  select(income_bracket, Math, Reading)
```

Question 18: Proportion of districts with a math gap

What proportion of districts in each income bracket have an average estimated math achievement gap favoring boys? Answer this question by performing the following steps:

- Append an indicator variable `math_gap_favoring_boys` to `seda_data_agg` that records whether the average estimated math gap favors boys *by more than 0.1 standard deviations relative to the national average.*
- Compute the proportion of districts in each income bracket for which the indicator is true: group by bracket and summarize `math_gap_favoring_boys` by taking the mean. Store the resulting data frame as `income_bracket_boys_favored`. What fraction of districts with a log income of $95,000-$119000 have a math gap favoring boys by more than 0.1 standard deviations?

```
# define indicator
income_bracket_boys_favored <- ...
# print result
```

Type your answer here, replacing this text.

Part 2: a simulation study

In lecture, we used the following code to simulate a population of red-tailed hawks:

```
set.seed(8888)

n_female <- 250
n_male <- 250

female_hawks <- tibble(
  length = rnorm(n = n_female, mean = 57.5, sd = 3),
  sex = "female"
)

male_hawks <- tibble(
  length = rnorm(n = n_male, mean = 50.5, sd = 3),
  sex = "male"
)

hawk_pop <- bind_rows(female_hawks, male_hawks) |>
  mutate(bird_id = str_c("H", sprintf("%03d", row_number()))) |>
  select(bird_id, sex, length)

num_hawks <- nrow(hawk_pop)
population_mean <- mean(hawk_pop$length)
```

We now use a simulation study to investigate how the variance of the sample mean (under simple random sampling with replacement) changes as a function of the sample size.

Question 19:

Complete the code below to approximate the variance of the sample mean for a given value of `sample_size`.

```r
set.seed(123)

sample_size <- 50
num_reps <- xxxxx
sample_means <- numeric(num_reps)

for (rep_index in 1:num_reps) {
  xxxxxx
  xxxxxx
}

var_sample_mean <- var(xxxxx)
```

Question 20:

We now approximate the variance of the sample mean for several values of `sample_size`. To make the code more organized and reusable, it is often a good idea to modularize it.

For this purpose, we will create a function `one_sample_size()` that takes a single sample size as input and returns the corresponding approximated variance.

```r
one_sample_size <- function(sample_size , num_reps = xxxxx){
  sample_means <- numeric(num_reps)

  for (rep_index in 1:num_reps) {
    xxxxxx
    xxxxxx
  }

  var_sample_mean <- var(xxxxxx)
  return(tibble(sample_size = sample_size,
                var_sample_mean = var_sample_mean))
}
```

Question 21:

Using the function defined above, approximate the variance of the sample mean for each of the given sample sizes.

```
sample_sizes <- c(25, 50, 100, 200)
all_results <- tibble()

for(sample_size in sample_sizes){
  one_sample_size_results <- xxxxxx
  all_results <- rbind(all_results,
                       one_sample_size_results)
}
```

Question 22:

Describe what quantities are plotted in the figure below.

Then think about a theoretical explanation for why the dots lie (almost) on a straight line.

(The second part is for your own reflection and is not required to be submitted.)

```
# you need to set eval = TRUE to run the ggplot() command
ggplot(all_results,
       aes(x = log10(sample_size), y = log10(var_sample_mean))) +
  geom_line() +
  geom_point() +
  theme_bw()
```